Optimization HW 1.

(Ref: lecture notes).

1(a) It is given that set: $C = C_1 \cap C_2$

C_1 and C_2 are convex sets.

Let $x, y \in C$

$\Rightarrow x, y \in C_1$ and $x, y \in C_2$

$x, y \in C_1$, and C_1 is convex set

1(a) $\Rightarrow \lambda x + (1-\lambda) y \in C_1$ for any $\lambda \in [0,1]$ — ①

$x, y \in C_2$, and C_2 is a convex set

$\Rightarrow \lambda x + (1-\lambda) y \in C_2$ for any $\lambda \in [0,1]$ — ②

From ① & ②

1(a) $\Rightarrow \lambda x + (1-\lambda) y \in C_1$ and C_2

$\lambda x + (1-\lambda) y \in (C_1 \cap C_2)$

Because $C_1 \cap C_2 = C$.

1(a) $\Rightarrow \lambda x + (1-\lambda) y \in C$

Given that $\lambda x + (1-\lambda) y \in C$

Implies C is a convex set.

(Ref: mostly lecture note)

1(b) let set $M_1 = \{(x, y) \in R^2 \mid x \leq 0, y \geq 0\}$

set $M_2 = \{(x, y) \in R^2 \mid x \geq 0, y \leq 0\}$

~~let~~ M_1 and M_2 convex set.

M_1 and M_2 constitute the points in 2nd and 4th quadrants in a 2-D plane (including the axes) respectively.

For $(x_1, y_1), (x_2, y_2) \in M_1$

1(b) and $(x_3, y_3), (x_4, y_4) \in M_2$

the convex set combination rule holds.

For any $(x_1, y_1), (x_2, y_2) \in M_1$, For any $\lambda \in [0, 1]$

$$\lambda(x_1, y_1) + (1-\lambda)(x_2, y_2) = (\lambda x_1 + (1-\lambda)x_2, \lambda y_1 + (1-\lambda) y_2)$$

It can be seen that

$$\left.\begin{array}{l} \lambda x_1 + (1-\lambda)x_2 \leq 0 \\ \lambda y_1 + (1-\lambda) y_2 \geq 0 \end{array}\right\} \in M_1 \text{ is convex}$$

Similarly M_2 is convex

$\therefore$ M_1 and M_2 are convex.

(Ref: lecture note)

1(b) Hence $M_1 \cup M_2$ is not convex even though M_1 and M_2 are convex.

Let us take $(-1, 2) \in M_1$
and $(2, -1) \in M_2$
and $\lambda \in [0, 1]$ let $\lambda = \frac{1}{2}$

$M_1 \cup M_2 = \{(-1, 2), (2, -1)\}$

$$\lambda x_1 + (1-\lambda) x_2 = \frac{1}{2}(-1) + \frac{1}{2}(2) = \frac{1}{2}$$
$$\lambda y_1 + (1-\lambda) y_2 = \frac{1}{2}(2) + \frac{1}{2}(-1) = \frac{1}{2}$$
$$\left(\frac{1}{2}, \frac{1}{2}\right) \notin M_1 \cup M_2$$

(b) $M_1 \cup M_2$ constitute point from 2nd and 4th quadrants in $\mathbb{R}^2$

$$\left(\frac{1}{2}, \frac{1}{2}\right) \notin (M_1 \cup M_2)$$

1(c) [Reference: 3.2.3 : Boyd / Vandenberghe]
Page # 80 Pg # 80.

$f_1 : R^n \to R$ $\quad$ $f_2 : R^n \to R$

Show that $f_{max}(x) = \max\{ f_1(x), f_2(x)\}$

Let domain of f = (domain f_1 ~~and~~ $\cap$ domain f_2)

domain of f = (domain f_1 $\cap$ domain f_2)

$(x, y) \in$ domain of f and $\theta \in [0, 1]$

$$f(\theta x + (1-\theta) y) = \max\{ f_1(\theta x + (1-\theta) y), f_2(\theta x + (1-\theta) y)\}$$

$$\leq \max\{(\theta f_1(x) + (1-\theta) f_1(y)), (\theta f_2(x) + (1-\theta) f_2(y))\}$$

When we take

$\theta f_1(x) + (1-\theta) f_1(y)$ this is $\leq \theta \max\{f_1(x), f_2(x)\} + (1-\theta)\max\{f_1(y), f_2(y)\}$

Similarly for $\theta f_2(x) + (1-\theta) f_2(y)$.

$$\theta f_2(x) + (1-\theta) f_2(y) \leq \theta \max\{ f_2(y) + f_2(x)\} + (1-\theta)\max\{f_1(y), f_2(y)\}$$

[Reference text Book: 3.2.3 Pg # 80]

Hence for

1(c)

$$f(\theta x + (1-\theta) y) \leq \max \{(\theta f_1(x) + (1-\theta) f_1(y)), (\theta f_2(x) + (1-\theta) f_2(y))\}$$

$$\leq \theta \max\{f_1(x), f_2(x)\} + (1-\theta) \max\{f_1(y), f_2(y)\}$$

$$= \theta f(x) + (1-\theta) f(y)$$

This follows as because each term is bounded by some quantity. Max of two number is bounded by common upper bound.

As shown above, This establishes convexity of $\underline{f}$.

2 $f_1 : \mathbb{R}^n \to \mathbb{R}$. $f_2 : \mathbb{R}^n \to \mathbb{R}$

2(a) $f_{min}(x) = \min\{f_1(x), f_2(x)\}$

Show that f_{min} is not convex

~~Lets~~ common e.g. of convex ~~functions~~

① linear :- from lecture (1).

$f_1(x) = x$ and $f_2(x) = -x$ are convex functions

$f_{min}(x) = \min\{f_1(x), f_2(x)\} = -|x|$

Let $\theta = \frac{1}{2}$, $x = -1$, $y = 1$

By definition of convexity :-

$$f_{min}(\theta x + (1-\theta)y) \leq \theta f_{min}(x) + (1-\theta) f_{min}(y)$$

$$f_{min}\left(\tfrac{1}{2}(-1) + \tfrac{1}{2}(1)\right) \leq \tfrac{1}{2} f_{min}(-1) + \tfrac{1}{2} f_{min}(1)$$

$$f_{min}(0) \leq \tfrac{1}{2} f_{min}(-1) + \tfrac{1}{2} f_{min}(1)$$

$$0 \leq -\tfrac{1}{2} - \tfrac{1}{2}$$

$$0 \leq -1/4 \xrightarrow{\text{FALSE}} \text{Not correct}$$

This serves as an e.g. that min of two convex fn is not convex

The earlier e.g. shows that

2(a) min of two convex fns is not convex

similarly Polynomial powers are convex function

↳ Lecture Rep

$f_1(x) = x^2 \qquad f_2(x) = (x-1)^2$

$$f_{min}(x) = \min\{f_1(x), f_2(x)\}$$
$$= \min\{x^2, (x-1)^2\}$$

By convex function def

$$f_{min}(\theta x + (1-\theta)y) \leq \theta f_{min}(x) + (1-\theta) f_{min}(y)$$

$$\theta = \frac{1}{2}, \quad x = 0, \quad y = 1$$

$$f_{min}\left(0 + \frac{1}{2}\right) \leq \frac{1}{2} f_{min}(0) + \frac{1}{2} f_{min}(1)$$

$$\frac{1}{4} \leq 0 + 0$$

$$\frac{1}{4} \leq 0 \quad \text{— ✗ FALSE.}$$

Hence min of the two convex functions $f_1(x) = x^2$, $f_2(x) = (x-1)^2$ is not a convex function

2(b) let $C_1 = \{(x, y) \mid y \leq 0\}$

C_1 is a closed half-space and Convex set

let $C_2 = \{(x, y) \mid y \geq e^{-x}\}$

The second derivative of e^{-x} is +ve

Hence e^{-x} is convex function

C_2 is the epigraph of continuous function e^{-x}

$\therefore C_2$ is closed and Convex set

[BOYD 2.5] page # 49 strict separation of two sets C and D must meet

$a^T x < b$ for all $x \in C$

and $a^T x > b$ for all $x \in D$

In this e.g $C_1 \cap C_2 = \{\} = \phi$

C_1 and C_2 are disjoint

However, as $x \to +\infty$, $e^{-x} \to 0+$

$\therefore$ Distance between C_1 and C_2 tends to zero.

$\longrightarrow$ Hence strict-separation condition is not met

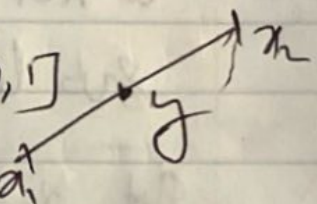
$\therefore C_1 = \{(x,y) \mid y \leq 0\}$, $C_2 = \{(x,y) \mid y \geq e^{-x}\}$ cannot be strictly seperated

2(c) [Reference: Lecture] 2.1 Convex Program

Sub-level sets of $f(x)$ are sets of the form:

$$L_c = \{ x \mid f(x) \leq c \}, \; c \in \mathbb{R}$$

Prove: If $f(x)$ is a convex function any of its sub-level sets are also convex



Let $x_1, x_2 \in L_c, \lambda \in [0,1]$

Let $y = \lambda x_1 + (1-\lambda) x_2$

$$f(y) = f(\lambda x_1 + (1-\lambda) x_2)$$

From definition of convexity

$$\leq \lambda f(x_1) + (1-\lambda) f(x_2)$$

$$\leq \lambda (c) + (1-\lambda) c$$

$$\boxed{f(y) \leq c} \Rightarrow y \in L_c$$

$\therefore$ Any sub level set of a convex function is a convex set.

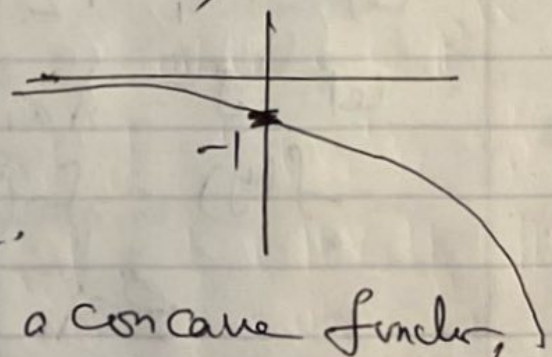
2(c) An e.g. of a function that is not convex but all of its sub-level sets are convex :

~~$f(x) = e^{-x}$~~

$$f(x) = -e^{x}$$

is not convex on $\mathbb{R}$, but all its sub level sets are convex

Plot of $f(x) = -e^{x}$



Here $-e^{x}$ is not convex.

Infact, concave. For a concave function, α- superlevel set, given by $\{x \in \text{dom } f \mid f(x) \geq \alpha\}$ is a convex set.

[Reference BOYD], 3.1.6.

sub level set: $L_c = \{ x \mid f(x) \leq c \}$

$$-e^{x} \leq c \Rightarrow e^{x} \geq -c$$

$$x \geq \ln(-c) \text{ for } c<0$$

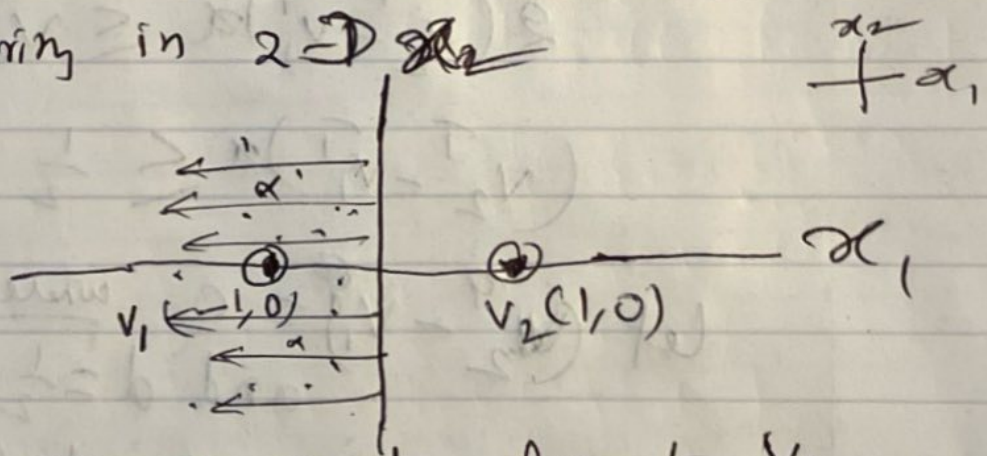
for $c<0$, $[\ln(-c), \infty]$ is convex

3a) Consider two points $v_1, v_2 \in R^n$

Show that $\exists$ a vector $c \in R^n$ and scalar $d \in R$

s.t. $\{x : \|x - v_1\|_2 \leq \|x - v_2\|_2\} = \{x : c^T x \leq d\}$

Considering in 2-D



Set of points closer to V_1 are in the region $x_1 \leq 0$.

$\Rightarrow$ (i.e. All the points to left of $x_1 = 0$) $(x_1 < 0)$.

3b) The shaded region corresponds to

$$\{(x_1, x_2)^T \mid x_1 \leq 0\}$$

Half-space - $\boxed{x_1 \leq 0}$

3 c

$$\{x : \|x - v_1\|_2 \leq \|x - v_2\|_2\}$$

$$(x - v_1)^T (x - v_1) \leq (x - v_2)^T (x - v_2)$$

$$\cancel{x^T x} - 2v_1^T x + v_1^T v_1 \leq \cancel{x^T x} - 2v_2^T x + v_2^T v_2$$

$$2(v_2^T - v_1^T)x \leq v_2^T v_2 - v_1^T v_1$$

$$(v_2^T - v_1^T)x \leq \frac{1}{2} v_2^T v_2 - v_1^T v_1$$

Let $(v_2^T - v_1^T) = c$ where $c \in \mathbb{R}^n$

and $d = \frac{1}{2}(v_2^T v_2 - v_1^T v_1)$

$\Rightarrow d = \frac{1}{2}(\|v_2\|_2 - \|v_1\|_2)$ where $d \in \mathbb{R}$

$$c^T x \leq d$$

$\Rightarrow \{x \in \mathbb{R}^n \mid c^T x \leq d\}$ is a half-space

i.e $\{x \in \mathbb{R}^n \mid cx \leq d\}$ is a half-space

Therefore, it follows that

$$\{x : \|x - v_1\|_2 \leq \|x - v_2\|_2\} = \{x \mid c^T x \leq d\}.$$